Gear Wall Element D

Our process for creating our conceptual designs was based on taking our design requirements from element C and turning them into functions. We had used these functions to make ideas for design aspects of the gear wall.

Morphological Chart

Function	Concept 1	Concept 2	Concept 3	Concept 4
Prevents harm to children	a clear plastic barrier in the front to enclose the gears/mechanism from the children	Rubber outer casing to cushion rough play from the kids and device getting harmed	the design has no sharp edges or corners to poke or pry the child	Uses non toxic materials to prevent poisoning
Engages with children	Led lights with different colors to catch attention while being able to adjust the brightness too	Different sounds playing depending on the design (car noise, piano sounds, chimes, etc)	Fast pace marbles moving around on different tracks against one another to reach the bottom first	Moving cars with different speeds depending on the sizes and shapes of the gears
Teaches cause and effect	Makes gears spin with different outcomes depending on the configuration of the gears	Cars moving at different speeds depending on gears and having them race each other	A conveyer that brings balls up, then rolls them down	A gear system using a dynamo DC Motor with it turning on depending on the amount of revolution of the crank
Improves motor skills	Incorporates a hand crank to spin the gears to do different objectives	Buttons you can press to adjust certain aspects such as different sounds and lights	Objects from varying shapes and sizes to push and pull to change obstacles or paths	A type of input that would require you to rotate a gear in a circle
Minimizes cost	Uses 3d printed gears and items	Uses wood at hand from the workshop	Use inexpensive materials	Simplify any complex mechanisms
Ways to be changed	Make gears removable and interchangeable	Make extensions to the gear wall that can be changed	Allow multiple input types for the same output or having multiple outcomes for the same input	Ability to change different colors and sound affect to the child's liking and to adjust the

				brightness/volume
Ways to prevent structure damage	Add structural support to the wall mounting bracket to reinforce the stability in the connection between the bracket and the device	The use of more stable/durable materials such as hard woods and metals will increase the structural integrity of the device	Add support to the more fragile pieces at multiple points to ensure stability in a directions to create longevity of the parts but also allowing easy access and interchangeability if broken	Make only what needs to be hollow on the inside hollow for stability
Unobtrusive	Make it foldable so it can be easily put away into storage	Make the gear wall easily detachable to move out of the way	Give the gear wall a design that will match the area around it to make it pop out less	Slim and sleek design to be compact and flush against the wall to prevent bulkiness and to prevent it from sticking out too much

Conceptual design Chart

	Concept 1-4			
Function	Plinko design	Cars conveyer	Dynamo light generator	Marble race track
Safe	soft edges with a clear plastic barrier to prevent fingers getting pinched from the gears and to remove access to the marbles.	soft edges with a clear plastic barrier to prevent fingers getting pinched from the gears and low speeds of the cars to prevent accidents	Low-voltage system, encased components with a transparent acrylic case	Rounded edges with a clear plastic barrier to enclose the entire contraption to prevent harm of the children and to prevent malfunctions from the kids
Engaging	Randomized marble descending with sound effects, lights and visual stimulation of the marbles falling to the bottom	Moving cars will engage the children. By giving children the ability to control conveyor speed and car path engagement can be increased.	The dynamo lights displaying some sort of “light show” when cranked	Fast-paced marbles moving around in all directions and spirals to compete one another to see which marble reaches the bottom first.

Conducive to Cause & effect Learning	Visual cause-effect as marble hits pegs as they descend and when the marbles fall into a specific column they receive a prize	Students can use controls to change the speed of the conveyor and the path which the car makes. This will provide the students exposure to cause and effect learning improving their skills.	By turning the crank, the gears turn on a LED display	Adjusting marble release position to affect the speed and the outcome of which th marble reaches the bottom
Accessibility	Able to adjust the height of the device with the movability of different components	The device could be fitted with interchangeable and easy to use cranks and knobs to allow accessibility by a diverse spectrum of users.	Ergonomic handles for students that may be limited with motor control	Multiple entry points and adjustable to different heights.
Fine motor skills	The ability to spin the crank to bring the marbles back to the top and being able to move the chute to adjust where the marbles falls	Using the control knobs will enhance the students fine motor skills and finger dexterity.	Turning the crank strengthens grip and control to turn the motor and using different handles can strengthen the different muscles such as using a ball shaped handle rather than a tube.	Being able to hold and place the marbles in different positions and having push/pull rods to adjust to different tracks
Cost effective	Made with basic pegs and accessible materials	Made with simple materials including wood and acrylic. By making the majority of the parts to reduce the cost.	The use of smaller motors in place of one big motor to power and to use more 3d printed parts	Making our own different tracks to reduce cost and to use materials already found in PW
Customizable	Able to change the light colors and different sound effect with the ability to move the shute to dispense the marbles	Handles, gears, and cars can be modified and exchanged to improve performance. Can be detached from the wall allowing	Interchangeable handle parts to crank the dynamo and the customization of different colors	Interchangeable track design to increase or decrease speeds of the marbles and to be able to change

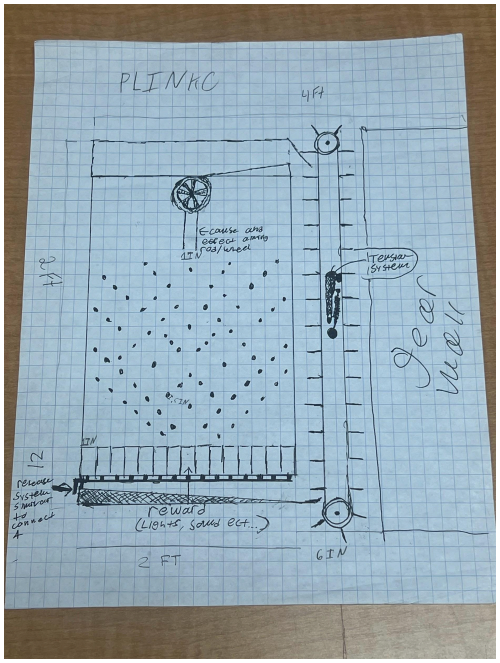
		devices to be interchanged to get the most out of the wall space.	depending on the child's preference.	different colors of the lights and marbles
Durable	Wood or sturdy plastic for the outer casing	Made with strong materials such as wood or acrylic. Designed in a manner which reduces wear on parts over time.	Its durability comes from a robust dynamo mechanism, encased to prevent damage, and strong materials like metal or hard plastic for the crank and housing.	Hard plastic or metal tracks for longevity
Unobtrusive	Slim design, wall-mounted and moverable	Does not have lights and will be designed in a manner as to not produce any annoying or distracting sounds.	With a compact, quiet system that fits neatly into a wall, blending seamlessly into any environment with a neutral or sleek appearance.	A slim wall-mounted design with the ability for it to moveable, so it can be changed out to different designs and the design will be painted to blend into the rooms aesthetics

Decision Matrix

1-5		Plinko Design		Cars Conveyor		Dynamo Light Generator		Marble Race Track	
Functions	Weight	Rating	Score	Rating	Score	Rating	Score	Rating	Score
Safe	15.00%	4	0.60	4	0.60	3.5	0.53	3.5	0.53
Engaging	15.00%	4.5	0.68	4	0.60	3.5	0.53	4.5	0.68
Cause & effect Learning	15.00%	3.5	0.53	4.5	0.68	4.5	0.68	4.5	0.68
Accessibility	15.00%	3	0.45	4	0.60	4	0.60	3.5	0.53
Fine motor skills	15.00%	4	0.60	3	0.45	4	0.60	3	0.45
Cost effective	6.67%	3	0.20	2.5	0.17	4	0.27	2.5	0.17
Customizable	6.67%	4.5	0.30	3	0.20	2.5	0.17	2.5	0.17
Durable	6.67%	3.5	0.23	2	0.13	3.5	0.23	2.5	0.17
Unobtrusive	2.50%	3.5	0.09	3.5	0.09	4	0.10	2.5	0.06
Maintainable	2.50%	3	0.08	3	0.08	3.5	0.09	2.5	0.06

Total	100.00%		3.75		3.59		3.78		3.48

Concept #1: Plinko Design



Description:
A wall-mounted interactive device where marbles descend through a peg-filled board, triggering lights and sounds to teach cause-and-effect. A crank resets the system, and a movable chute allows control over marble release. Durable, safe, and customizable, it supports motor skill development and fits into various settings.

Advantages:

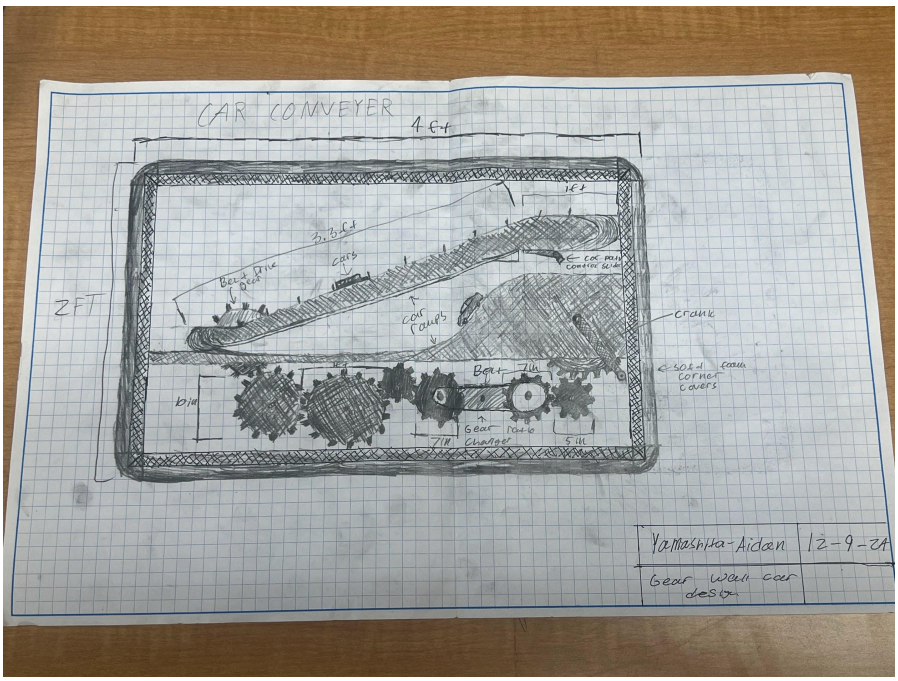
- **Safe**: Clear barriers prevent access to mechanisms and marbles, reducing hazards.
- **Engaging**: Randomized paths, lights, and sounds provide sensory stimulation.
- **Educational**: Demonstrates cause-and-effect relationships.
- **Accessible**: Adjustable height and movable components suit diverse users.

- **Motor Skills**: Crank and chute movement improve strength and coordination.
- **Cost-Effective**: Made with basic, affordable materials.
- **Customizable**: Options for colors, sounds, and chute positions.
- **Durable**: Sturdy materials ensure longevity.
- **Compact**: Slim, wall-mounted, and movable design.

Disadvantages:

- Crank operation may challenge some users.
- Limited interactivity beyond marble drop and crank use.
- Requires maintenance for optimal function.
- Customization features add to initial costs.
- Needs sufficient wall space.

Concept #2: Cars Conveyor



Description:
An interactive device where children control moving cars on a conveyor belt, exploring cause-and-effect with adjustable speed and path controls. Constructed with safe, durable materials like wood and acrylic, it encourages motor skill development and is accessible for various abilities.

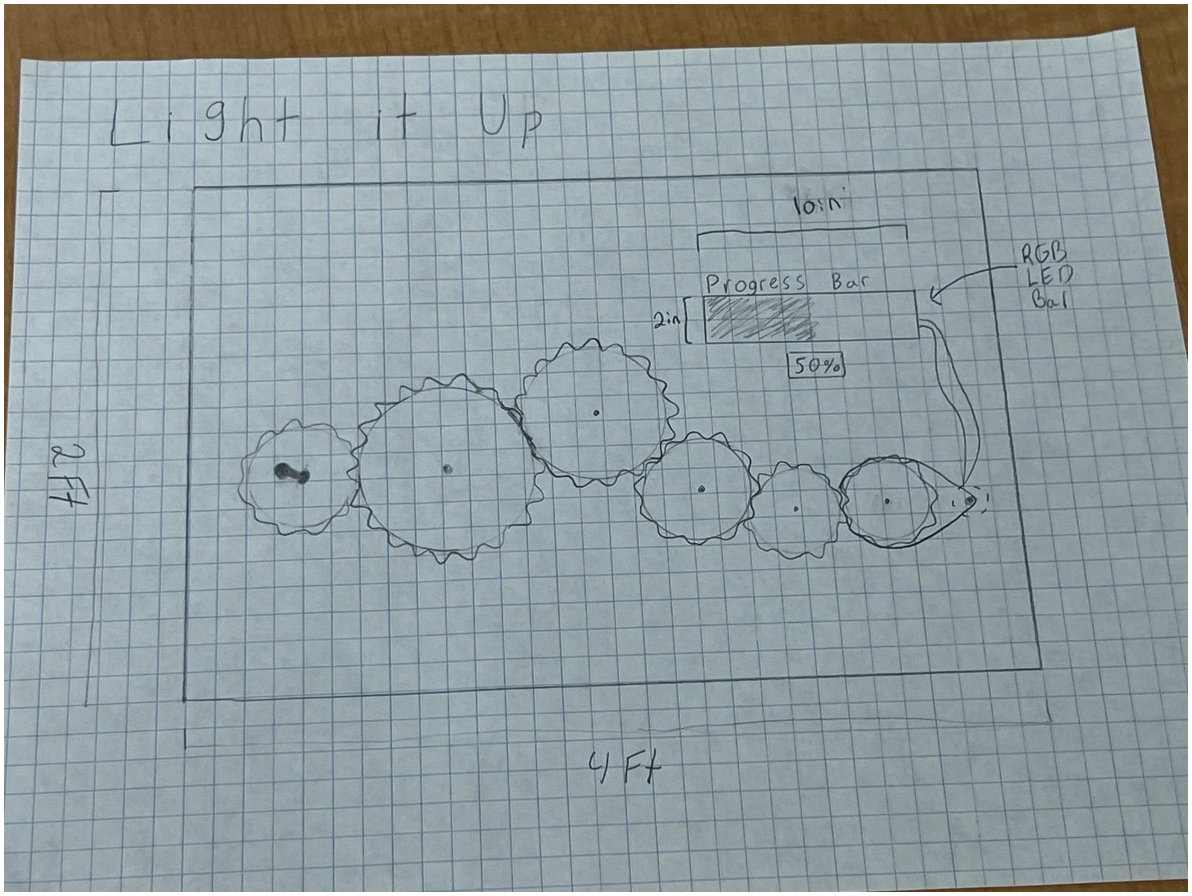
Advantages:

- **Safe**: Soft edges and clear barriers protect against injury.
- **Engaging**: Moving cars and control options capture attention and invite experimentation.
- **Educational**: Adjusting controls demonstrates cause-and-effect relationships.
- **Accessible**: Interchangeable cranks and knobs suit diverse users.
- **Motor Skills**: Controls improve hand-eye coordination and fine motor skills.
- **Cost-Effective**: Simple materials and in-house production lower costs.
- **Customizable**: Changeable components allow for user-specific adjustments.
- **Durable**: Built to withstand frequent use with sturdy materials.
- **Unobtrusive**: Quiet design fits into focused environments.

Disadvantages:

- **Limited Interaction**: Interaction is mainly limited to controlling car movement, which may not engage all children.
- **Wear and Tear**: Gears and moving parts may wear with frequent use.
- **Space Needs**: Requires wall space for installation and operation.
- **Limited Controls**: Might lack complexity for children seeking more challenges.
- **Customization Costs**: Custom parts may increase initial production costs.

Concept #3: Dynamo Light Generator



Description:
A crank-powered device that activates LED lights, teaching cause-and-effect and improving motor skills. Enclosed in a safe acrylic case, it features ergonomic handles and a compact design for classrooms.

Advantages:

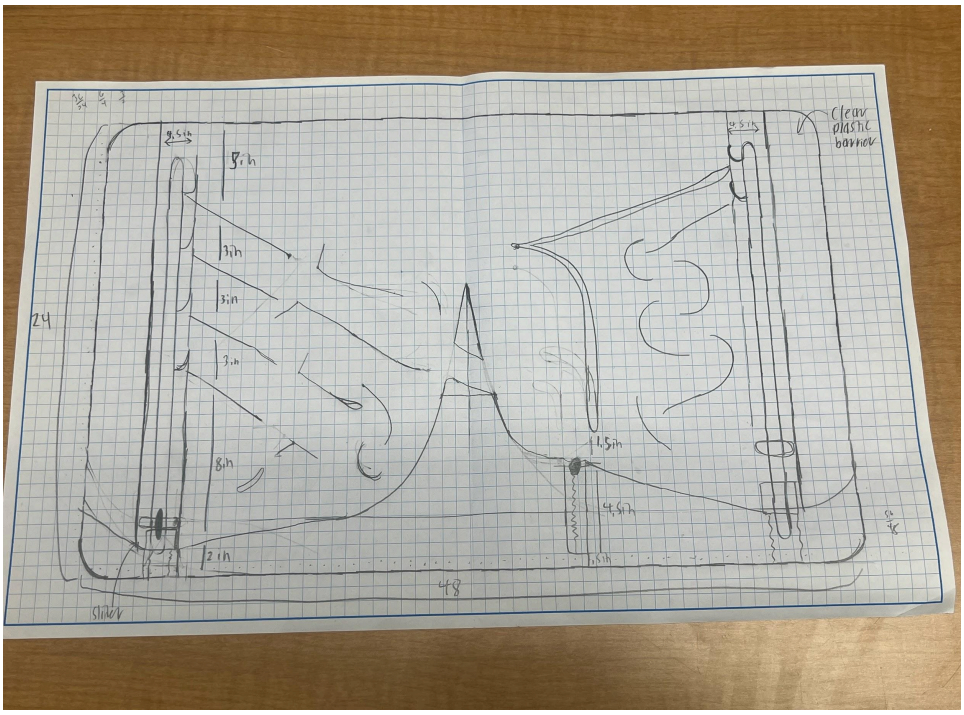
- **Safe**: Low-voltage and enclosed parts ensure user safety.
- **Engaging**: LED light displays promote interaction.
- **Educational**: Immediate feedback reinforces cause-and-effect learning.
- **Accessible**: Ergonomic handles accommodate varied abilities.
- **Motor Skills**: Strengthens muscles through handle use.
- **Cost-Effective**: Uses small motors and 3D-printed parts.
- **Customizable**: Changeable handles and LED colors.
- **Durable**: Built with sturdy materials.
- **Compact**: Quiet and space-efficient design.

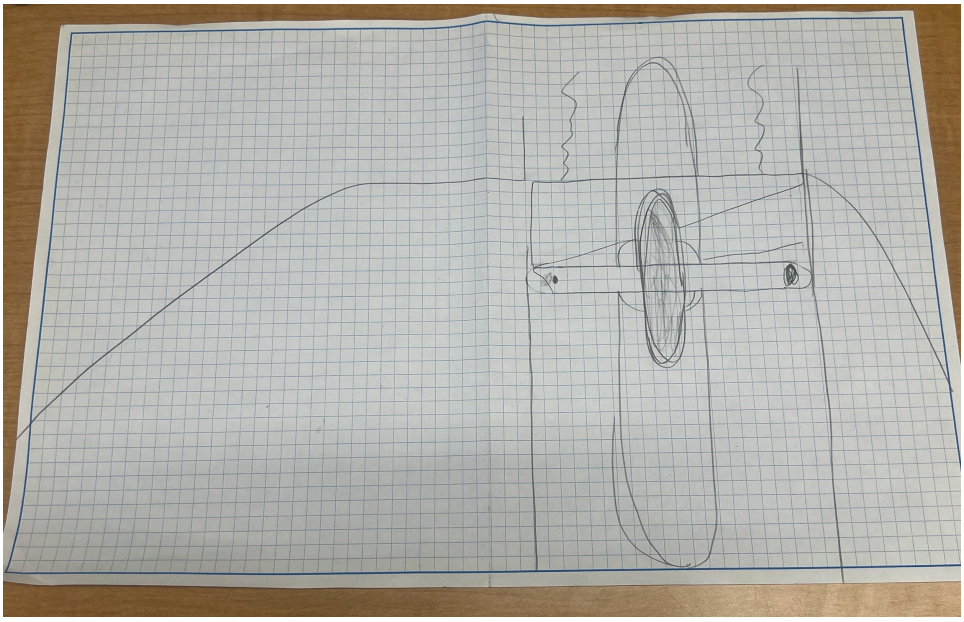
Disadvantages:

- Interchangeable parts increase assembly complexity.

- Small motors may limit LED brightness.
- Relies entirely on user-generated power.
- Higher upfront costs due to durable materials.
- Limited to crank-based interaction.

Concept #4: Marble Race Track





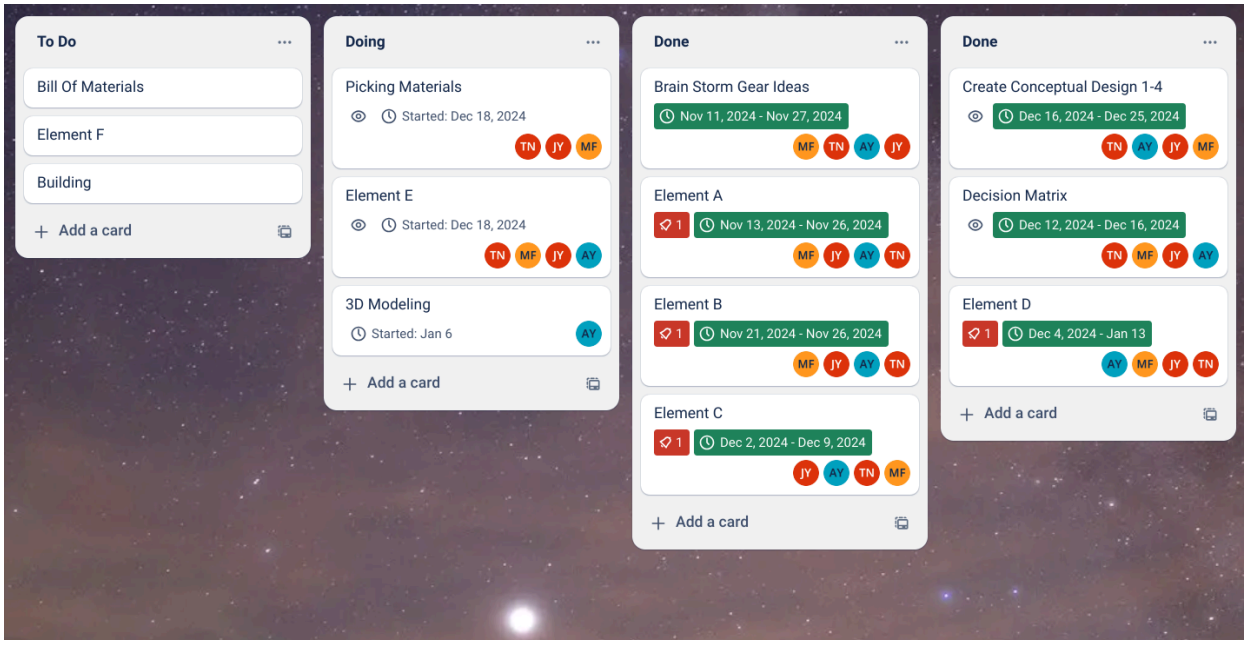
Description:
A wall-mounted track system where marbles race through loops and drops, encouraging motor skill development and competitive play. Accessible, durable, and customizable, it suits diverse users and environments.

Advantages:

- **Safe**: Clear barriers and rounded edges prevent injuries.
- **Engaging**: Competitive, visually exciting design.
- **Educational**: Demonstrates how placement impacts speed and outcomes.
- **Accessible**: Adjustable height and multiple entry points.
- **Motor Skills**: Improves precision and coordination.
- **Cost-Effective**: Uses recycled or local materials.
- **Customizable**: Interchangeable tracks and design options.
- **Durable**: Sturdy construction for frequent use.
- **Compact**: Slim design fits various spaces.

Disadvantages:

- Limited interaction beyond marble racing.
- Tracks may require cleaning and alignment.
- Customization adds to initial costs.
- Heavy reliance on user action.



Our Rationale:

Based on our decision matrix, our group chose our conceptual design #3, “Dynamo Light Generator.” We selected this design because it offers simplicity, practicality, and a strong alignment with our project goals. The crank-powered mechanism effectively balances engagement and educational value, providing immediate feedback through LED light activation. This reinforces the concept of cause-and-effect while fostering motor skill development in users.

The design prioritizes safety with its low-voltage system and enclosed components, making it ideal for use in diverse educational settings. Its compact, durable build ensures longevity and ease of integration into classroom environments. Furthermore, the customizable features, such as interchangeable handles and LED colors, allow the device to be tailored to various user needs and preferences, enhancing accessibility.

While the design has some limitations, such as the complexity of assembly due to interchangeable parts and dependence on user-generated power, these drawbacks are outweighed by the advantages. The use of cost-effective materials like small motors and 3D-printed components helps manage production costs, and the emphasis on ergonomic design ensures usability across a wide range of abilities.

Overall, the “Dynamo Light Generator” provides a well-rounded solution that meets the educational, interactive, and safety requirements of our target audience, making it the best choice for our project.